LESSON 2.8

Describe Data Using the Mean

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can determine the mean of a data set, use a target mean to find a missing value, and explain the mean as the fair share and the balance point of the data.

WHY THIS MATTERS Every week, Miguel walks dogs after school to earn money, and he keeps his earnings in a notebook. Across 4 days he earned $60 in all — but not the same amount each day. Later this lesson you'll help Sofia, who tracks her points every basketball game and wants to finish the season averaging 15 points per game.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Mean *Spanish: Media*	The average of a data set. Add all the values, then divide by how many values there are.	Add, then divide by the count
Average *Spanish: Promedio*	Another word for the mean — the amount each value would be if the total were shared equally.	A typical amount
Measure of center *Spanish: Medida de centro*	A single number used to describe the middle, or typical value, of a whole data set.	Mean and median are both
Balance point *Spanish: Punto de equilibrio*	The value where the total distance of the data above it equals the total distance below it. The mean is the balance point.	The mean balances the set
Data set *Spanish: Conjunto de datos*	The whole collection of values that were gathered or measured.	8, 15, 11, 6
Target mean *Spanish: Media objetivo*	The mean a person is trying to reach.	A goal average of 29 points.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Four friends have 3, 5, 6 and 10 stickers. Find the mean.

1. Add: 3 + 5 + 6 + 10 = 24.
2. Count: there are 4 values.
3. Divide: 24 ÷ 4 = 6. Check: 6 is between 3 and 10.

Answer: 6 stickers each

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New data set: a dog walker earns $9, $11, $16, and $12 over 4 days. First, what is the total?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The total of all the values is _____ .
- There are _____ values, so I divided by _____ .
- The mean is _____ , which means _____ .
- The mean must fall between _____ and _____ .

Word bank mean | average | total | sum | divide | count | fair share | balance point | center | typical | data set | value

WATCH OUT!

A common mistake

The most common mistake with the mean is adding correctly and then dividing by the wrong number

LESSON 2.8

Describe Data Using the Mean

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Put the steps for finding the mean in the correct order.

Groups: Steps to find the mean (in order)

_____ Step 1: Add every value in the data set to find the total

_____ Step 2: Count how many values are in the data set

_____ Step 3: Divide the total by the number of values

_____ Step 4: Check that the answer lands between the smallest and largest value

HINT Think about the counters: first you combine them all, and only then do you share them out.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Find the mean of: 4, 6, 8, 10

(A) 6
(B) 7
(C) 8
(D) 28

HINT The mean is what each value would be if the total were shared out equally among all four numbers.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

3 **Circle the letter of the best answer. Show how you know.**

Miguel earned $60 walking dogs over 4 days. What is his mean daily earning?

(A) $4
(B) $15
(C) $20
(D) $60

HINT A typical day is the amount Miguel would earn if he earned the SAME amount all 4 days.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Find the mean of: 3, 5, 7, 9, 11

(A) 5
(B) 7
(C) 9
(D) 35

HINT Mean means average: level all five numbers out so they are all the same.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

5 **Circle the letter of the best answer. Show how you know.**

In her first 4 games, Sofia scored 12, 20, 16, and 11 points. What is her mean points per game so far?

(A) 14.75
(B) 15
(C) 20
(D) 59

HINT The mean here answers: how many points per game has Sofia averaged so far?

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

The Mean Mistake (forgot to divide)

1. **Data set** Points scored in 4 games: 8, 10, 6, 12
2. **Goal** Find the MEAN number of points per game.
3. **Add the values** 8 + 10 + 6 + 12 = 36
4. **Answer** Mean = 36 points

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** The mean levels the total out across the games — a total by itself has not been leveled yet.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Four games are below 15 and four games are above 15. Does that already mean the data balances at 15? What else would you have to measure?

> **SENTENCE STARTER** Counting games is not enough because ___. To balance, I have to measure how ___ each score is from ___, and then compare the total ___ to the total ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.8

Describe Data Using the Mean

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table.

Data Set	Total	How Many Values	Mean
6, 9, 12, 9			
20, 14, 16, 18, 22			
5, 8, 8, 11, 13, 15			

SHOW YOUR WORK

2 **Mark and label each value on the number line.**

Data: 10, 12, 14, 16, 18. Place a marker at the mean of this data set.



SHOW YOUR THINKING

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Sofia wants to average 15 points per game over 10 games. How many points does she need in all 10 games combined?

(A) 1.5 points

(B) 15 points

(C) 25 points

(D) 150 points

PER 1

4 **Circle the letter of the best answer. Show how you know.**

Sofia's points in games 1 through 9 were 12, 20, 16, 11, 17, 14, 15, 10, and 18. How many points has she scored so far?

(A) 15 points

(B) 123 points

(C) 133 points

(D) 150 points

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Sofia has 133 points after 9 games and needs 150 points in all to average 15 over 10 games. How many points must she score in game 10?

(A) 13 points

(B) 15 points

(C) 17 points

(D) 283 points

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

The Wrong Divisor

1. **Data set** Points in 4 games: 6, 11, 9, 14
2. **Goal** Find the MEAN number of points per game.
3. **Add the values** 6 + 11 + 9 + 14 = 40
4. **Divide** The student divides by 5: 40 ÷ 5 = 8, so the mean is 8.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Four games are below 15 and four games are above 15. Does that already mean the data balances at 15? What else would you have to measure?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.8

Describe Data Using the Mean

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table.

Game	Points Scored	Points Above or Below the Target Mean of 15
Game 1	12	
Game 2	20	
Game 4	11	
Game 7	15	

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A data set has a mean of 20. Every value in the data set is then increased by 4. What is the new mean?

(A) 5

(B) 20

(C) 24

(D) 80

Put the data in order _____

WORK

PART 3 Explain and correct

reasoning

3 **Answer in complete sentences.**

Sofia realized she made an error: she actually scored 12 points in Game 8, not 10. Her other games are unchanged (12, 20, 16, 11, 17, 14, 15, and 18). How does this change the number of points she needs in Game 10 to average 15 points per game? Show how you know.

WRITE YOUR ANSWER

4 **Find the mistake. Explain it, then show the correct work.**

The Missing Zero

1. **Data set** Goals allowed in 4 games: 0, 3, 2, 7
2. **Goal** Find the MEAN number of goals allowed per game.
3. **Add the values** 0 + 3 + 2 + 7 = 12
4. **Divide** The student says the shutout does not count as a game, so 12 ÷ 3 = 4 goals per game.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Four games are below 15 and four games are above 15. Does that already mean the data balances at 15? What else would you have to measure?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help